

# Lava Client — User Manual

A task-by-task guide to the Lava Android client, derived from the actual feature modules and screens in the repository. UI labels quoted below are taken from the real Compose screens / string resources (source files named per section). Anything not verifiable from the repo is marked UNCONFIRMED:.

Classification: project-specific.

## What Lava is

Lava is an unofficial Android client for popular Russian torrent trackers. It ships with a pluggable Tracker SDK that supports multiple content providers (see [Supported providers](#)). You search and browse trackers, view topic detail and comments, download .torrent files or open magnet links, and manage favorites, bookmarks, and history — all from one app. (Source: [README.md](#).)

A companion **Lava API** service does the actual scraping and exposes a JSON API to the app. You can connect the client to a Lava API on your local network, a cloud/remote server, or one running **on this very device** — see [On-device API](#).

## Supported providers

Six providers are wired through the Tracker SDK. Verified from each provider's TrackerDescriptor under core/tracker/<id>/:

| Provider<br>(display name) | Tracker id | Auth type     | Notes                                                                |
|----------------------------|------------|---------------|----------------------------------------------------------------------|
| RuTracker.org              | rutracker  | CAPTCHA_LOGIN | Login + captcha (full capability set incl. forum, favorites, upload) |
| RuTor.info                 | rutor      | FORM_LOGIN    | Anonymous by default                                                 |
| Internet Archive           | archiveorg | NONE          | Anonymous — no credentials needed                                    |
| Project Gutenberg          | gutenberg  | NONE          | Anonymous — no credentials needed                                    |
| Kinozal.tv                 | kinozal    | FORM_LOGIN    | Credentials                                                          |
| NNM-Club                   | nnmclub    | FORM_LOGIN    | Credentials                                                          |

Auth types come from `AuthType (core/tracker/api/.../AuthType.kt)`: `NONE`, `FORM_LOGIN`, `CAPTCHA_LOGIN`, `OAuth`, `API_KEY`. A `NONE` provider is always anonymous; a `FORM_LOGIN/CAPTCHA_LOGIN` provider that supports `Anonymous` shows an anonymous toggle, otherwise it requires credentials (`TrackerDescriptor.kt` KDoc table).

UNCONFIRMED: the `README.md` "Supported trackers" table lists only `RuTracker` + `RuTor` (it dates to 1.2.0). The six descriptors above are present in the source tree today; the `README` table is stale relative to them.

## Install / setup

Lava is distributed to testers via **Firebase App Distribution** (the project's distribution channel per the §6.P / §6.Z release process and the distribute logs under `.lava-ci-evidence/distribute-changelog/`). You receive a Firebase invitation, accept it, and install the APK from the link.

UNCONFIRMED: `README.md` also shows Google Play / GitHub / RuStore badges for the client app. The active, repo-verified distribution path is Firebase App Distribution; whether the public store listings are live is not determinable from the repository.

The client comes in two variants you can install side by side: a dev/debug build with the `.dev` application-id suffix and a release build. (Source: root `CLAUDE.md` commands section.)

## On-device API install (no Play Store)

If you choose to run the API **on this device**, the client links to the separate **Lava API app**. From onboarding's "On this device" section (or **Settings** → **Run the API on this device**), the button reads **"Install Lava API app"** when it is not installed and **"Open Lava API app"** when it is (source: `feature/onboarding/.../steps/ApiSelectionStep.kt`, lines 204-210). When not installed, the client uses a **Firebase download fallback** — **there are no dead Play-Store links** (source: `docs/CONTINUATION.md` linking-feature entry, 2026-06-04).

## Onboarding

The onboarding flow is a stepped wizard (source: `feature/onboarding/.../OnboardingScreen.kt`, steps `Welcome` → `ApiSelection` → `Providers` → `Configure` → `Summary`).

1. **Welcome** — brand logo, the heading **"Welcome to Lava"**, a "`N` providers available" line, and a **"Get Started"** button. (Source: `WelcomeStep.kt`.)
2. **Choose your API** (`ApiSelectionStep`) — heading **"Choose your API"** with three sections (source: `ApiSelectionStep.kt`):
  - **On your network** — auto-discovers Lava APIs over mDNS. Shows "Searching for APIs on your network..." while scanning, then "Found `N` API(s):" or "No APIs discovered on your network." Tap a discovered entry to run a connectivity probe; on success onboarding advances, on failure it

- shows "Could not reach this API: ..." with a **"Try again"** button. A **"Search again"** button rescans.
- **On this device** — **"Install Lava API app"** / **"Open Lava API app"** (see above).
- **Cloud / remote server** — pick a preset or type a server address into the **"Server address (https://host:port)"** field and tap **"Add server"**.
- 3. **Pick your providers** (ProvidersStep) — heading **"Pick your providers"**, "Select one or more content providers to configure." Each provider row shows a colored dot, its display name, its auth type (e.g. "Form Login"), and a checkbox. The **"Next"** button is enabled once at least one is selected. (Source: ProvidersStep.kt.)
- 4. **Configure** — per-provider configuration (credentials / anonymous toggle). See [Provider configuration](#).
- 5. **Summary** — review and finish.

## Provider configuration

Source: [feature/provider\\_config/.../ProviderConfigScreen.kt](#).

The provider-config screen composes sections including a **Sync** section and — only when `descriptor.supportsAnonymous == true` — an **Anonymous** section toggle. Credentials for login-based providers are entered through the credentials / login screens (see [Login](#)).

## Search

Source: [feature/search\\_input/.../SearchInputScreen.kt](#), [feature/search\\_result/.../SearchResultScreen.kt](#).

1. Open the search input. It shows a search field (the hint comes from the `designsystem designsystem_hint_search` string) with a clear (×) action.
2. A **provider chip bar** (ProviderChipBar) lets you toggle which configured providers the search runs against. (Source: SearchInputScreen.kt lines 92-94.)
3. Submit the query to see results in SearchResultScreen. Results carry a **filter** affordance and a category row that starts with **"All"** followed by per-category chips. (Source: SearchResultScreen.kt.)
4. If a provider requires login for search, instead of a misleading "Nothing found" the results show a **Login required** empty-state with a **Login** button (SP-3.2 change; source comment in SearchResultScreen.kt lines 280-293).

## Browse / categories

Source: [feature/category/.../CategoryScreen.kt](#), [feature/forum/.../ForumScreen.kt](#), [feature/search\\_result/.../categories/CategorySelectionScreen.kt](#).

Browse the tracker's category / forum tree. Category and forum entries render by name. If a forum requires login to browse/search, a **"Login required"** dialog appears with **Login** / cancel actions (source: CategoryScreen.kt lines 258-271).

## View a topic + comments + download

Source: [feature/topic/.../TopicScreen.kt](#) and [feature/topic/src/main/res/values/strings.xml](#).

The topic screen shows the topic title, poster image, and content, with a share action and a paginated comments list. Verified actions:

- **Magnet** button (`topic_action_magnet = "Magnet"`) — opens a magnet dialog with share / open / cancel actions. Shown only when the topic carries a non-blank magnet link.
- **Torrent** button (`topic_action_torrent = "Torrent"`) — downloads the .torrent file. If you are not authorized, a **"Login required"** dialog appears: *"To download the torrent file you need to be authorized. If you do not have an account you can use a magnet link."* (`topics_login_required_for_download`).
- **Download** requires storage permission; if not granted, a rationale dialog **"Grant access to storage?"** is shown (`permission_write_storage_rationale_title`).
- **Add a comment** — opens an "Add a comment" dialog with a "Write your message..." placeholder and send / cancel actions (`topic_add_comment_title / topic_add_comment_placeholder`). Writing comments requires authorization.
- A favorite toggle is available on the topic.

## Login

Source: [feature/login/.../LoginScreen.kt](#), [feature/login/.../ProviderLoginScreen.kt](#).

For providers with `FORM_LOGIN` or `CAPTCHA_LOGIN` auth, the login screens let you enter credentials (and, for RuTracker, solve a captcha). Anonymous-capable providers can be used without credentials. Credentials are stored securely and never logged (per §6.H — see [docs/security/README.md](#)).

UNCONFIRMED: exact on-screen field labels for the login screens were not read line-by-line in this pass; the behavioral contract (form login vs. captcha vs. anonymous) is derived from the descriptors and `AuthType`.

## Favorites, bookmarks, history

Sources: [feature/favorites/.../FavoritesScreen.kt](#), [feature/bookmarks/.../BookmarksScreen.kt](#), [feature/visited/](#).

- **Favorites** — topics you have favorited (toggled from the topic screen).
- **Bookmarks** — bookmarked forums/categories.
- **History (visited)** — recently visited items.

Each of these has its own periodic **sync** setting in the menu (see below), with configurable sync periods.

## Settings menu

Source: [feature/menu/.../MenuScreen.kt](#)

- `feature/menu/src/main/res/values/strings.xml`. Verified rows include:
- **Settings** section: **Theme** selector; the connection/endpoint selection item; **"Run the API on this device"** row; **"Provider Credentials"**; per-feature sync periods (Favorites / Bookmarks / History / Credentials).
- **Data** section: clear history, clear bookmarks, clear favorites.
- **Misc** section: rights, privacy, contacts, **About**.
- A **Sign out** confirmation flow.

## On-device API

The client can connect to a Lava API running on your own phone/tablet. See the dedicated guide: [docs/guides/ON\\_DEVICE\\_API\\_USER\\_GUIDE.md](#) (start/stop/restart, the access key/pairing, mDNS discovery, the persistent notification, privacy notes, and its own troubleshooting table). Not duplicated here.

## TV support

Lava ships a `TvActivity` (leanback) entry point alongside the phone/tablet `MainActivity` (source: `app/src/main/AndroidManifest.xml`, `<activity android:name=".TvActivity">` with `LEANBACK_LAUNCHER`). The manifest declares `android.software.leanback` and `android.hardware.touchscreen` as **not required**, so the app installs on TV-class devices.

## Deep links

The manifest registers `rutracker.org` deep links for `viewtopic.php`, `viewforum.php`, and `tracker.php` (both `http` and `https`), so tapping a RuTracker URL can open it directly in Lava (source: `app/src/main/AndroidManifest.xml` `MainActivity` intent filter).

## Troubleshooting

| Symptom                                             | What to check                                                                                                                                                                                                                                                    |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Onboarding can't find an API ("No APIs discovered") | Ensure a Lava API is running on the same Wi-Fi and your router allows multicast; tap <b>"Search again"</b> . Or use <b>Cloud / remote server</b> (enter <code>https://host:port</code> ) or <b>On this device</b> . (Source: <code>ApiSelectionStep.kt</code> .) |
| "Could not reach this API" after selecting one      | The connectivity probe failed; tap <b>"Try again"</b> or pick a different API.                                                                                                                                                                                   |

| Symptom                                      | What to check                                                                                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Search shows <b>Login required</b>           | The selected provider needs credentials for search/download — tap <b>Login</b> and sign in (RuTracker also needs a captcha).                            |
| .torrent download blocked / "Login required" | Authorize the provider, or use the <b>Magnet</b> link instead (RuTracker download requires login).                                                      |
| "Grant access to storage?" before download   | Grant the storage permission so the .torrent can be written to your download folder.                                                                    |
| Provider missing from the search chip bar    | Configure/enable it in onboarding or <b>Provider Credentials</b> ; only configured providers appear.                                                    |
| Cloud API returns 401                        | The cloud server must register your client build's identity. (Source: docs/CONTINUATION.md "cloud-search auth lag" entry — a known operator-side step.) |

## See also

- Security posture: [docs/security/README.md](#)
- On-device API user guide: [docs/guides/ON\\_DEVICE\\_API\\_USER\\_GUIDE.md](#)
- SDK developer guide (adding a provider): [docs/sdk-developer-guide.md](#)
- Architecture: [docs/ARCHITECTURE.md](#)
- Local network discovery: [docs/LOCAL\\_NETWORK\\_DISCOVERY.md](#)